SYSTEMS
MANAGEMENT



OBMF Specifications and Workstream Status Update

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OCP PLATINUM Member

Agenda

- OCP OBMF Workstream Updates
  - Workstream Intro
  - Status of specifications
  - Plans
- Delivered Content
- OBMF-ICP deployment models
- OBMF-ICP Ready products and source code
- Call to action

OBMF Workstream

- OBMF - Open Boot and Management Framework
- Closed CLA Workstream under HW Mgmnt WG by: Microsoft, Intel, AMD, ASPEED, Arm, Nuvoton, Nvidia, Renesas, Google, HPE, Dell, Microchip, AMI, Lattice Semiconductor, Code Construct, Marvell, NXP, Renesas
- Weekly WS meetings since Q2 2025

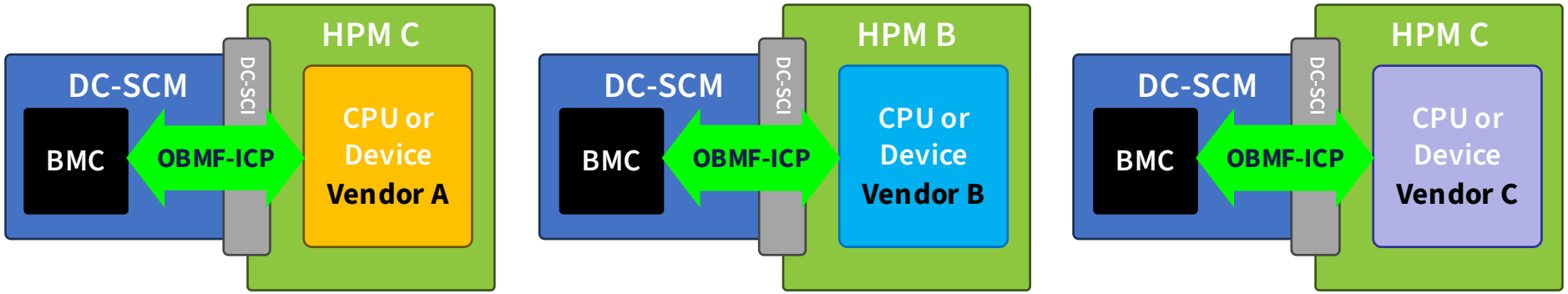
Scope:

A specification to prescribe the DC-SCM interface (or a list of recommended physical interfaces) and standardized protocol for use in boot, configuration and runtime communication between Host Processor or other devices on the Host Processor Module (HPM) and Baseboard Management Controller (BMC) on the DC-SCM Module.

OBMF - Open Boot and Management Framework

Definition of OBMF Interface Consolidation Protocol (OBMF-ICP)

- Vendor Agnostic and CPU/Device Agnostic (PCIe Card, CXL Card, AMC, etc.)
- Physical Interface Agnostic (Protocol layer definition)
- Aligned to DC-SCM 2.x and Multi Node in DC-SCM 3.0
- Generic and extensible for Vendor Adoption (self describing extensions and abstractions)
- Leveraging OCP (e.g., Streaming Boot) & DMTF Standards (e.g., MCTP)



OBMF Specifications - Status



OBMF-ICP (Interface Consolidation Protocol) specification revision 1.0.0 released

OBMF-ICP over USB revision 1.0 released (see 2026 EMEA Summit Announcement)

OBMF-ICP over MCTP: DSP0295 WIP document released by DMTF
(https://www.dmtf.org/sites/default/files/DSP0295_1.0.0WIP80.pdf)

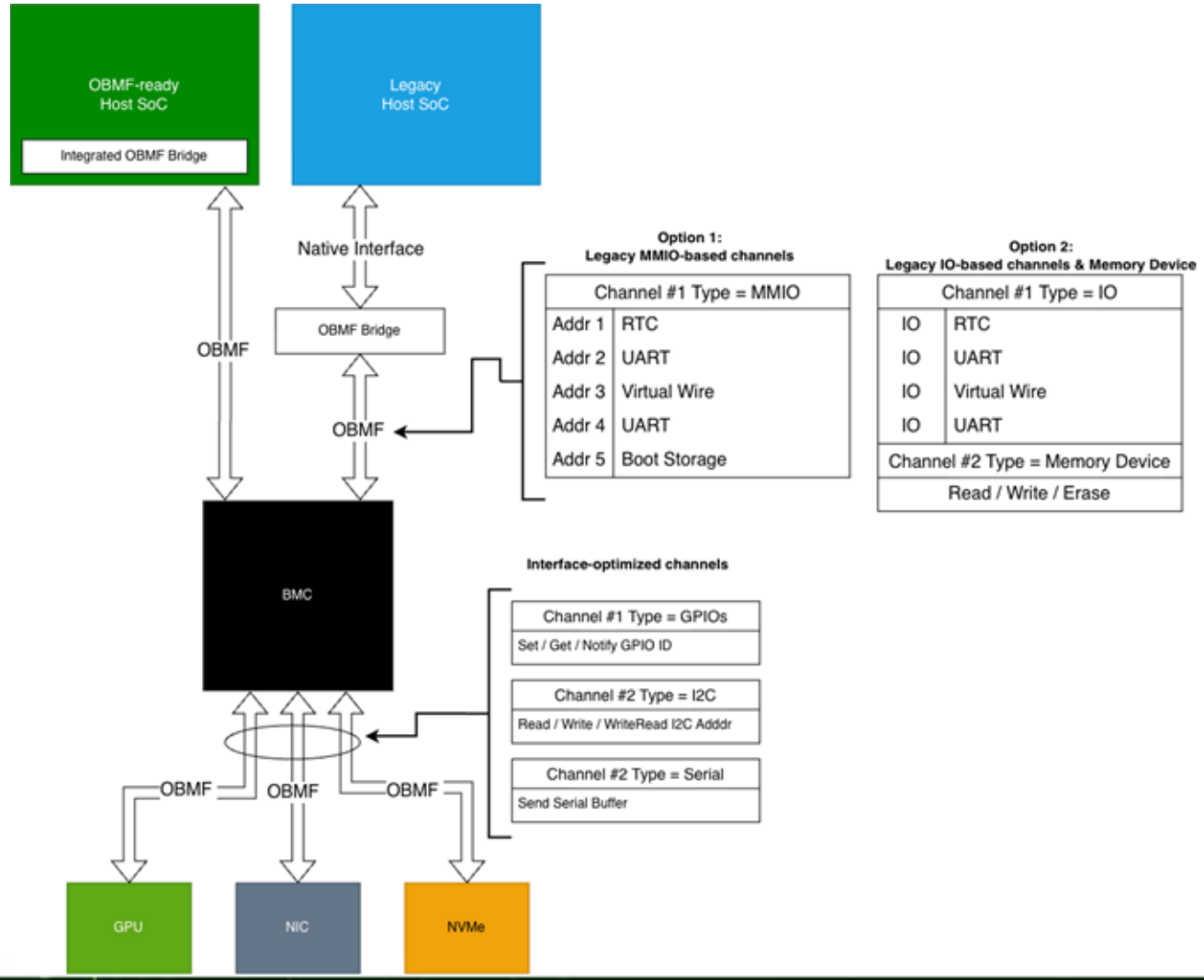
OBMF Workstream influenced PNP and DC-SCM work

OBMF WS Plans

- Platform topology discovery mechanisms - in cooperation with PNP and DC-SCM OCP workstreams
- Releasing OBMF-ICP over MCTP via DMTF.org
- OBMF-ICP WIP (revision 1.1 or later) to cover:
  - New use cases:
    - BIOS/OS driven reconfiguration
    - Security considerations
  - New OBMF-ICP Optimized Channels for:
    - Physical devices: PWM, TACH, ADC
    - Physical interfaces: JTAG Initiator, MCTP PCIe VDM traffic initiator, eSPI, SPI Multi-transaction support
    - Protocols: TPM, KVM (Keyboard Video and Mouse), Redfish host interface

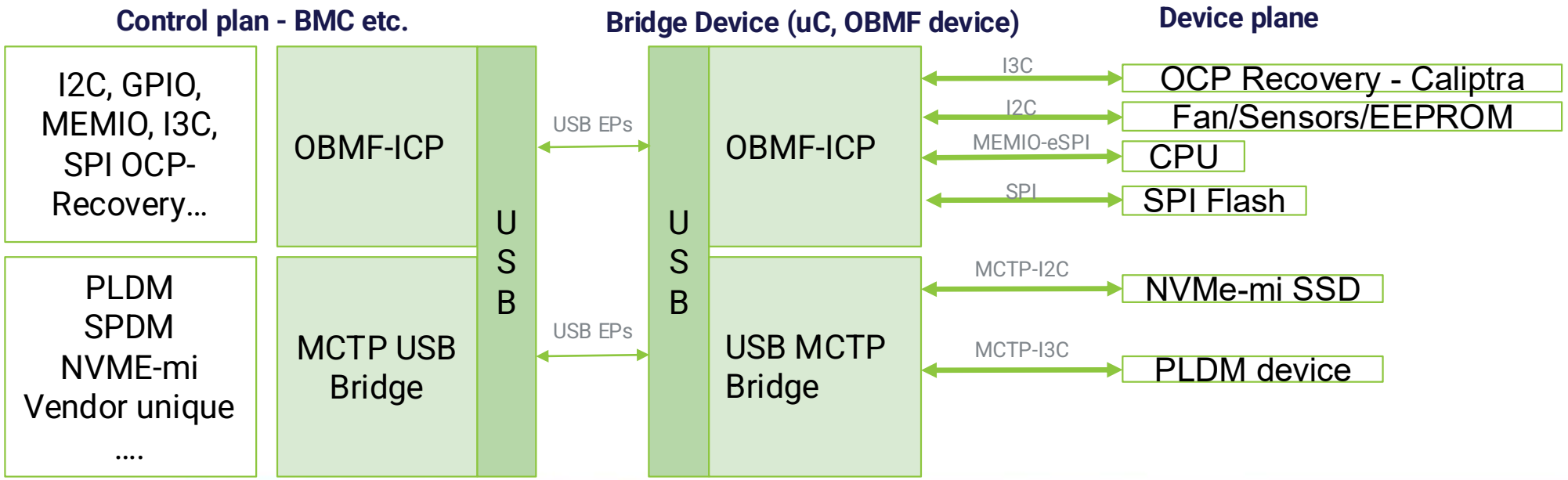
OBMF-ICP rev 1.0 Specification

- **Specification published here:**
  https://www.opencompute.org/documents/obmf-icp-1-0-0-pdf
- **Major platform use cases**
  - **Host SoC to BMC (OS/BIOS/SoC FW)**
  - **BMC to Devices (GPU/NIC/VR/sensors)**
- **Supported OBMF-ICP Channels**
  - **MMIO and I/O channels**
  - **Controllers: GPIO, SPI, I2C, I3C**
  - **Protocols: UART, IPMI, Memory Device**



Protocol Interoperability

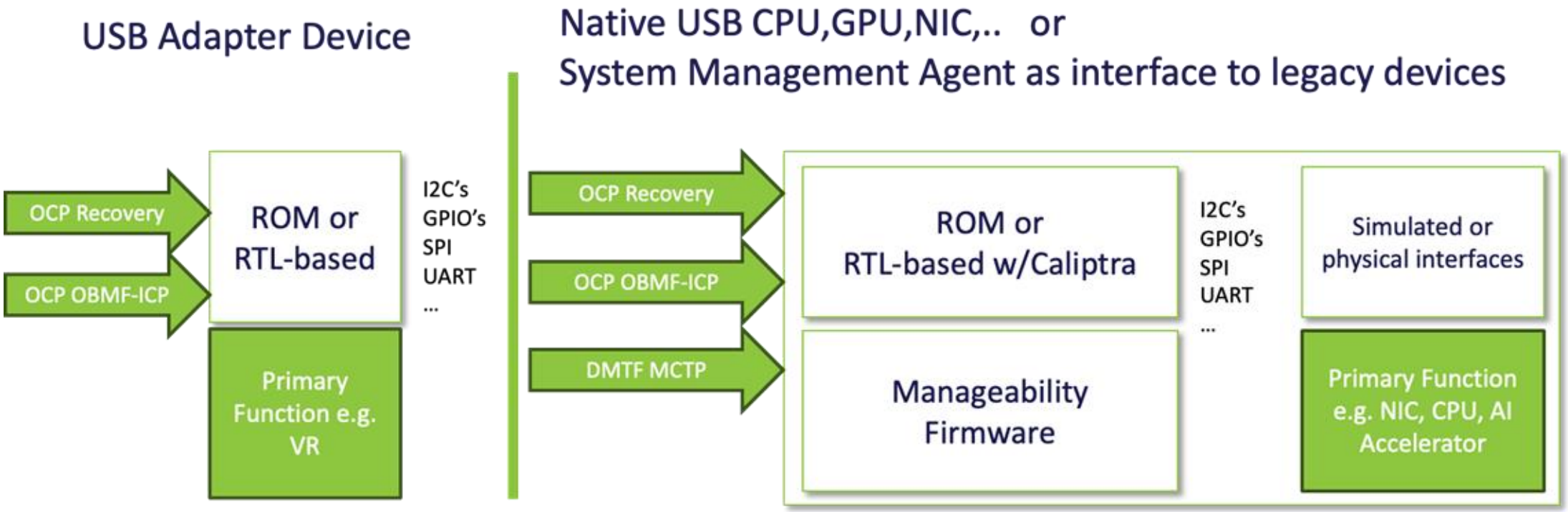
- Mix protocols on the same bus: OCP Recovery, OBMF-ICP, and MCTP (USB binding - DSP0283) on different USB EPs.
- MCTP co-exist with OBMF-ICP
- OBMF-ICP over MCTP - optionality



OBMF-ICP Deployment models

Definition of OBMF-ICP Specification together with OBMF-ICP USB Class definition is closing the last gap in unified USB Device definition

The following classes of USB devices can be now easily integrated into Datacenter



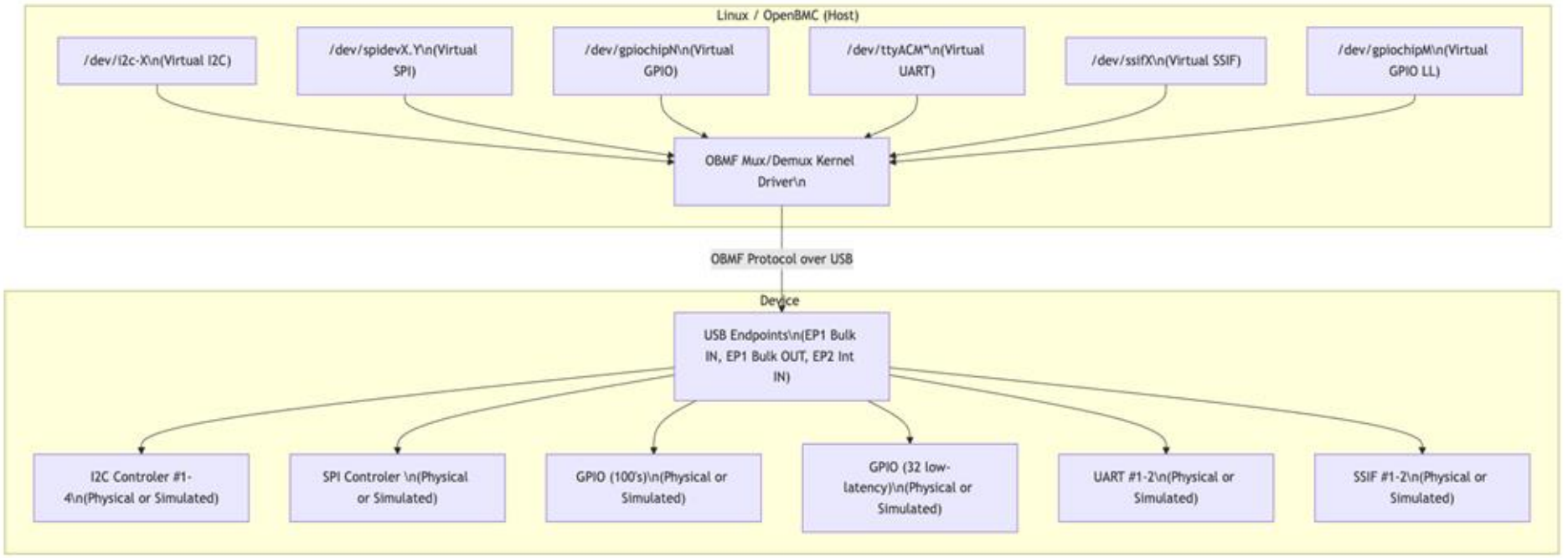
OBMF-ICP Optimized Channels Definition

FW-oriented Channel Optimization Principles:

- Maximize FW efficiency (minimize FW overhead, maximize BW usage)
- Maximize FW reuse (no changes to services and applications)
- Minimize FW changes (align to existing Linux driver frameworks)

BMC

OBMF Device



OBMF-ICP Ready products and source code

AMD – AMD's EPYC Reference Platform with Aspeed 27xx implements OBMF host and AMD's Instinct MI350P accelerator implements OBMF device

Arm - OBMF-ICP on the Arm Neoverse Fixed Virtual Platform (FVP) simulation model, including open source proof-of-concept with platform firmware, OpenBMC and OpenSMA patches available at https://gitlab.arm.com/server_management/PoCs/fvp-poc/-/tree/main/docs/obmf

ASPEED - AST2700 (BMC) and AST1040 (SMC/AMC) provide the OBMF ecosystem with a validated development platform aligned with the OBMF-ICP across various channels such as SPI, I2C, GPIO UART and many more on both upstream and downstream from BMC

Code Construct - developed a platform-neutral, open-source OBMF-ICP protocol implementation, for both host- and device-side components, which provides a straightforward foundation for porting to ICP management hardware designs

Intel – Intel platform support for OBMF-ICP with FPGA-based and SMC/AMC-based PoC implementations - enabling contributions to OpenBMC/OpenSMA and additional use cases

Marvell - product support for OBMF-ICP announcement aligned with OCP Global Summit

NVIDIA - please take a look at reuse for OBMF-ICP of:

- https://github.com/NVIDIA/lstp-qa **test suite,**
- https://github.com/NVIDIA/lstp_module **Linux driver**
- https://github.com/NVIDIA/OpenSMA/ **source code for MCU with USB infrastructure**

Nuvoton - Nuvoton's management controllers support OBMF-ICP, enabling standardized platform management, and showcase the solution at the OCP Global Summit

NXP - OBMF-ICP Bridge Zephyr stack running on NXP MCUs such as MCX-N9xx and MCX-A5xx, along with the Linux stack running on MPUs such as i.MX93 and i.MX95

Call to Action

- Review published OBMF specifications, provide feedback, join the OBMF WS!
- Participate in external standardization organizations to help OBMF
  - OCP DC-SCM 3.x, OCP Security, & OCP MHS PnP
  - DMTF.org
- Use OBMF-ICP over USB in your products, plan to use OBMF-ICP over MCTP
- See OBMF wiki page for up-to-date info: publication list, how to join, etc.
  https://github.com/opencomputeproject/ocp-obmf/wiki

Visit Innovation Village to see 7 (seven!) OBMF demos in a common area

Open Discussion

